

Parth Patel

+1 669-235-6091 | patelsparth00@gmail.com | linkedin.com/in/parth-patel-sjsu | github.com/parthpatel00 | patelparth.me

WORK EXPERIENCE

AI Engineer, SJSU Research Foundation, San Jose, CA

May 2025 – May 2026

- Deployed a fine-tuned model as a containerized FastAPI service with Docker across multiple GPU nodes, keeping code review responses under 3 seconds each even when an entire class submits in the same hour.
- Fine-tuned an open-source LLM on 20,000+ graded assignment submissions with PyTorch so it reviews student code the way course staff do, giving 600+ students at 2 universities feedback the day they submit.

Software Engineer, Deloitte, Montreal, QC

Jul 2022 – Oct 2024

Client: Public Health Agency of Canada

- Created a data validation microservice in Python on AWS (SQS, Lambda, DynamoDB) that screens 20K+ monthly vaccine submissions for schema and completeness errors, cutting invalid records reaching provincial pipelines by 94%.
- Introduced Docker-based CI/CD pipelines on AWS with automated build checks and deployment monitoring, cutting release time for the 5-province public health platform from 2 hours to about 30 minutes.
- Migrated 2M+ immunization case records from on-prem Oracle to AWS RDS with 100+ SQL scripts that remap legacy schemas onto the new cloud data model, landing the full move with zero data loss.
- Implemented Python Azure Functions that accept FHIR health records from 5 provinces, flatten the nested payloads, and unify them in Azure Cosmos DB, making new cases queryable in minutes rather than overnight.
- Instrumented the ingestion pipeline with CloudWatch dashboards and alerts that flag queue backups and error spikes, cutting the time to notice a failed submission from hours to under 5 minutes.

Client: Employment and Social Development Canada

- Delivered a GPT-3.5 RAG chatbot that answers caseworker policy questions from a 10,000+ document knowledge base in Azure Cosmos DB and cites its sources, so a policy lookup that took about 15 minutes now takes under 30 seconds.
- Replaced a 30-minute manual handoff with a CI/CD pipeline that compiles the React/TypeScript chatbot and pushes the production bundle straight to ESDC's repository, letting the team deploy in one step.
- Scripted the chatbot's monthly knowledge base refresh with Python and bash jobs that pull updated policy documents, re-chunk them, and rebuild the search index overnight, replacing about 4 hours of manual updates each cycle.

PROJECTS

TenMunches – SF Food Discovery Platform | Python, AWS (Lambda, RDS, S3, EC2), DynamoDB, React, TypeScript, Node.js, Vercel

- Launched TenMunches, a live food discovery site that ranks San Francisco restaurants by what reviewers actually say rather than raw star averages, now used by 400+ people to pick where to eat.

QuizMaster – AI Quiz Generator | Python, AWS (Lambda, S3, DynamoDB, API Gateway), React, TypeScript, Gemini API, Vercel

- Shipped QuizMaster, a live web app that turns any uploaded PDF, whether lecture notes or a training manual, into a personalized quiz in under 5 seconds so students can drill from their own material.
- Ran the entire backend serverless on AWS Lambda, S3, and API Gateway, keeping hosting costs near zero at idle while absorbing traffic spikes when a class finds the app before finals.

ACHIEVEMENTS

- Earned Deloitte's Silver Dot award for the delivery of the Public Health Agency of Canada's vaccine surveillance platform, an honor given to a handful of practitioners across the practice each cycle.
- Won 4 hackathons in 6 months: CalHacks 12.0 (2,100 participants), Hack For Humanity '26, and 2 tracks at SJHacks '26.

EDUCATION

San Jose State University, MS in Software Engineering (AI Specialization), GPA: 3.65

Jan 2025 – May 2026

University of Toronto, BS in Mathematics and Statistics

Sep 2018 – Jun 2022

Relevant Coursework: Cloud Technologies, Distributed Systems, Machine Learning, Deep Learning, Database Systems.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Bash, TypeScript, HTML, CSS

Databases: Oracle, DynamoDB, Cosmos DB

Cloud/Frameworks: AWS (Lambda, S3, SQS, RDS, EC2, CloudWatch), Azure (Functions), GitHub Actions, Docker, Git, FastAPI, React

Certifications: Google Cloud Professional Data Engineer, Google Cloud Associate Cloud Engineer, Microsoft Azure Data Engineer Associate (DP-203), Microsoft Azure Fundamentals (AZ-900), Microsoft Azure Data Fundamentals (DP-900)